SE **Lab – 6**

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Code and Branch Coverage Analysis

L6_L7_37_irrigation_scheduler

Code and Branch Coverage Analysis

1. Aim

To test the Irrigation Scheduler Python application using Pytest and measure code coverage and branch coverage using Coverage.py.

2. Procedure

1. Created test cases for the irrigation scheduler.
2. Executed the test cases using Pytest.
3. Measured code coverage using Coverage.py.
4. Enabled branch coverage and generated the final report.
5. Checked the test results and coverage percentages.

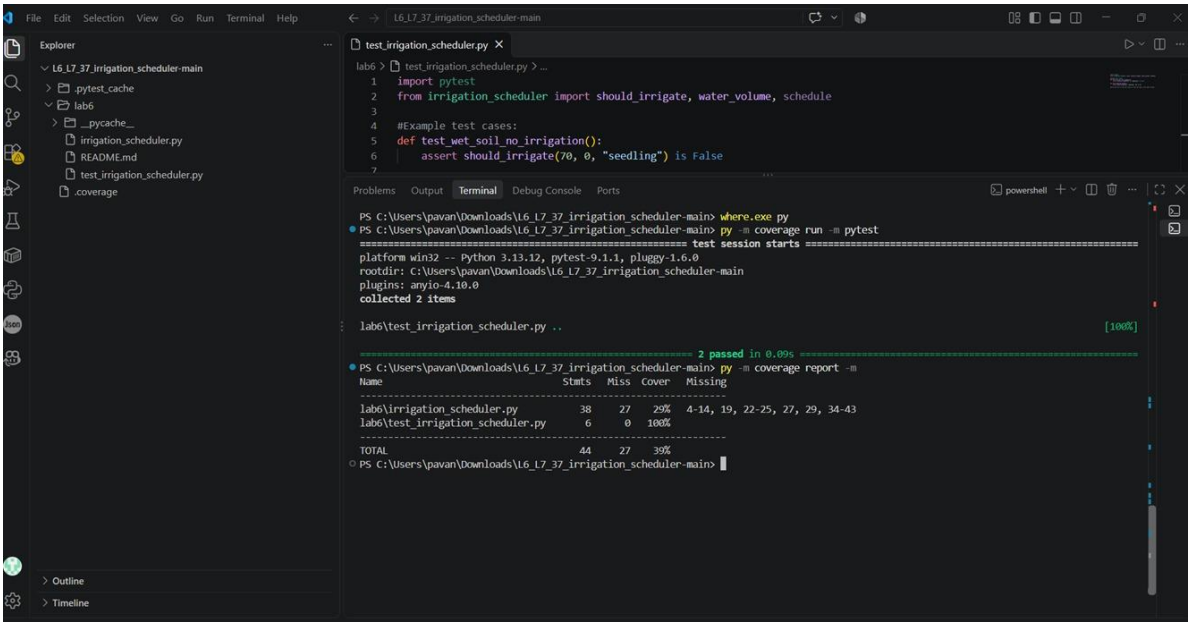
3. Results

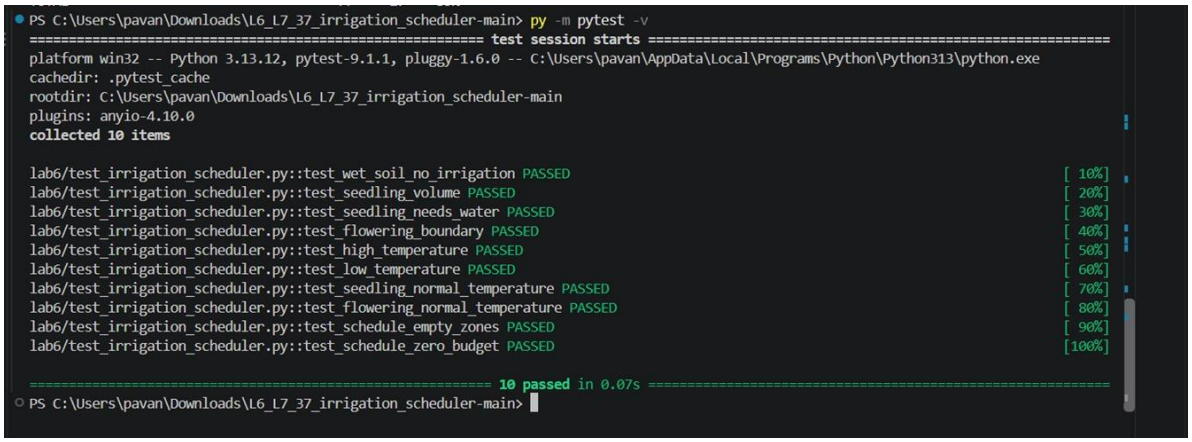
- **Total test cases: 18**
- **Tests passed: 18**
- **Tests failed: 0**
- **Statement coverage: 100%**
- **Total statements: 80**
- **Missed statements: 0**
- **Branches reported: 28**
- **Partial branches: 0**
- **Total coverage in the branch-enabled report: 100%**

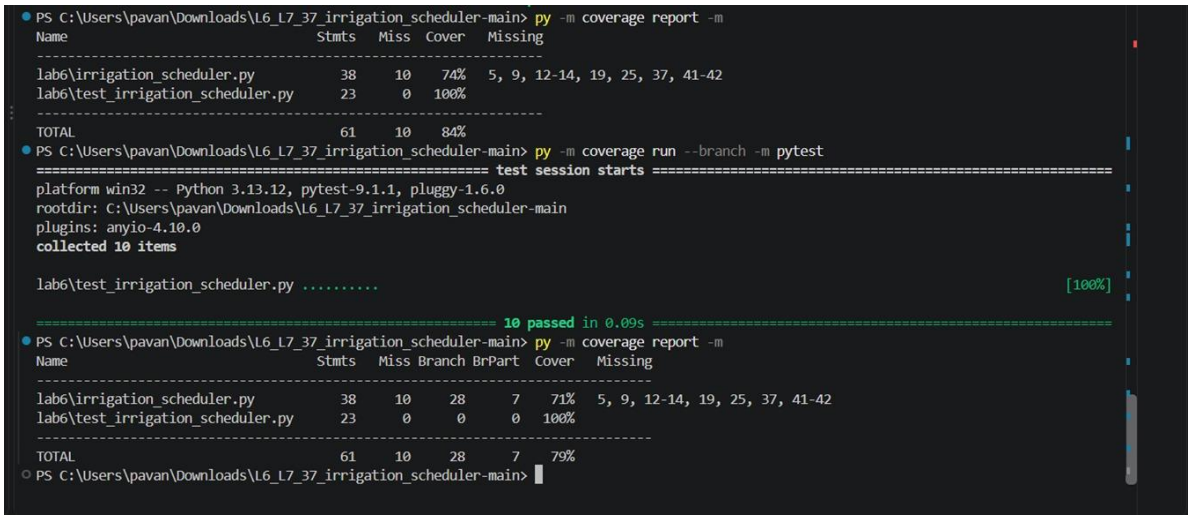
4. Conclusion

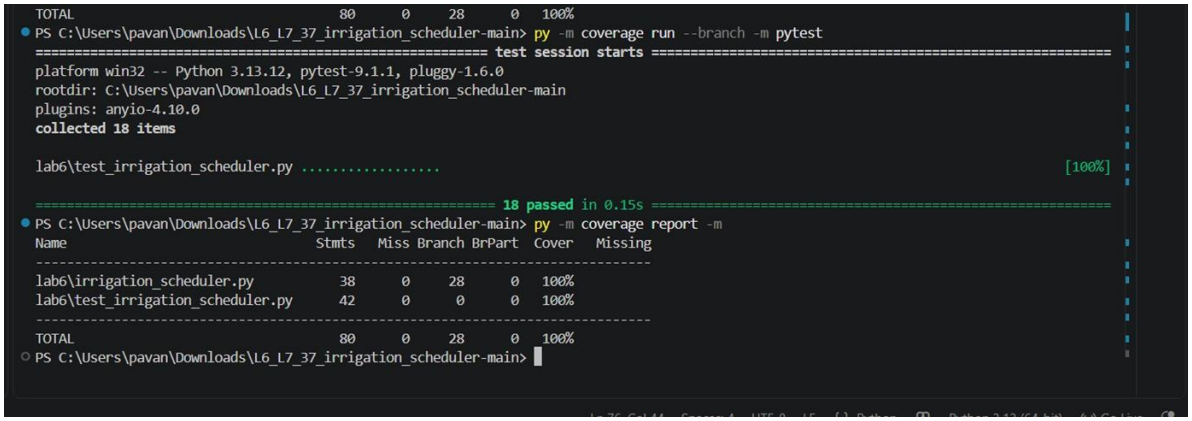
All 18 test cases passed successfully. The coverage report showed 100% statement coverage and 100% total coverage in the branch-enabled report. This indicates that the measured statements and branch outcomes were covered by the tests.

1.


File Edit Selection View Go Run Terminal Help
L6_L7_37_irrigation_scheduler-main
Explorer
L6_L7_37_irrigation_scheduler-main
.pytest_cache
lab6
__pycache__
irrigation_scheduler.py
README.md
test_irrigation_scheduler.py
.coverage
test_irrigation_scheduler.py
lab6 > test_irrigation_scheduler.py > ...
1 import pytest
2 from irrigation_scheduler import should_irrigate, water_volume, schedule
3
4 #Example test cases:
5 def test_wet_soil_no_irrigation():
6 assert should_irrigate(70, 0, "seedling") is False
Problems Output Terminal Debug Console Ports
powershell
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> where.exe py
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage run -m pytest
test session starts
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main
plugins: anyio-4.10.0
collected 2 items
lab6\test_irrigation_scheduler.py .. [100%]
2 passed in 0.09s
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage report -m
Name Stmts Miss Cover Missing
lab6\irrigation_scheduler.py 38 27 29% 4-14, 19, 22-25, 27, 29, 34-43
lab6\test_irrigation_scheduler.py 6 0 100%
TOTAL 44 27 39%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>
Outline
Timeline


PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m pytest -v
test session starts
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\pavan\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main
plugins: anyio-4.10.0
collected 10 items
lab6/test_irrigation_scheduler.py::test_wet_soil_no_irrigation PASSED [10%]
lab6/test_irrigation_scheduler.py::test_seedling_volume PASSED [20%]
lab6/test_irrigation_scheduler.py::test_seedling_needs_water PASSED [30%]
lab6/test_irrigation_scheduler.py::test_flowering_boundary PASSED [40%]
lab6/test_irrigation_scheduler.py::test_high_temperature PASSED [50%]
lab6/test_irrigation_scheduler.py::test_low_temperature PASSED [60%]
lab6/test_irrigation_scheduler.py::test_seedling_normal_temperature PASSED [70%]
lab6/test_irrigation_scheduler.py::test_flowering_normal_temperature PASSED [80%]
lab6/test_irrigation_scheduler.py::test_schedule_empty_zones PASSED [90%]
lab6/test_irrigation_scheduler.py::test_schedule_zero_budget PASSED [100%]
10 passed in 0.07s
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>


PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage report -m
Name Stmts Miss Cover Missing
lab6\irrigation_scheduler.py 38 10 74% 5, 9, 12-14, 19, 25, 37, 41-42
lab6\test_irrigation_scheduler.py 23 0 100%
TOTAL 61 10 84%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage run --branch -m pytest
test session starts
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main
plugins: anyio-4.10.0
collected 10 items
lab6\test_irrigation_scheduler.py .... [100%]
10 passed in 0.09s
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage report -m
Name Stmts Miss Branch BrPart Cover Missing
lab6\irrigation_scheduler.py 38 10 28 7 71% 5, 9, 12-14, 19, 25, 37, 41-42
lab6\test_irrigation_scheduler.py 23 0 0 0 100%
TOTAL 61 10 28 7 79%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>


TOTAL 80 0 28 0 100%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage run --branch -m pytest
test session starts
platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main
plugins: anyio-4.10.0
collected 18 items
lab6\test_irrigation_scheduler.py ..... [100%]
18 passed in 0.15s
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py -m coverage report -m
Name Stmts Miss Branch BrPart Cover Missing
lab6\irrigation_scheduler.py 38 0 28 0 100%
lab6\test_irrigation_scheduler.py 42 0 0 0 100%
TOTAL 80 0 28 0 100%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>